

DataSheet

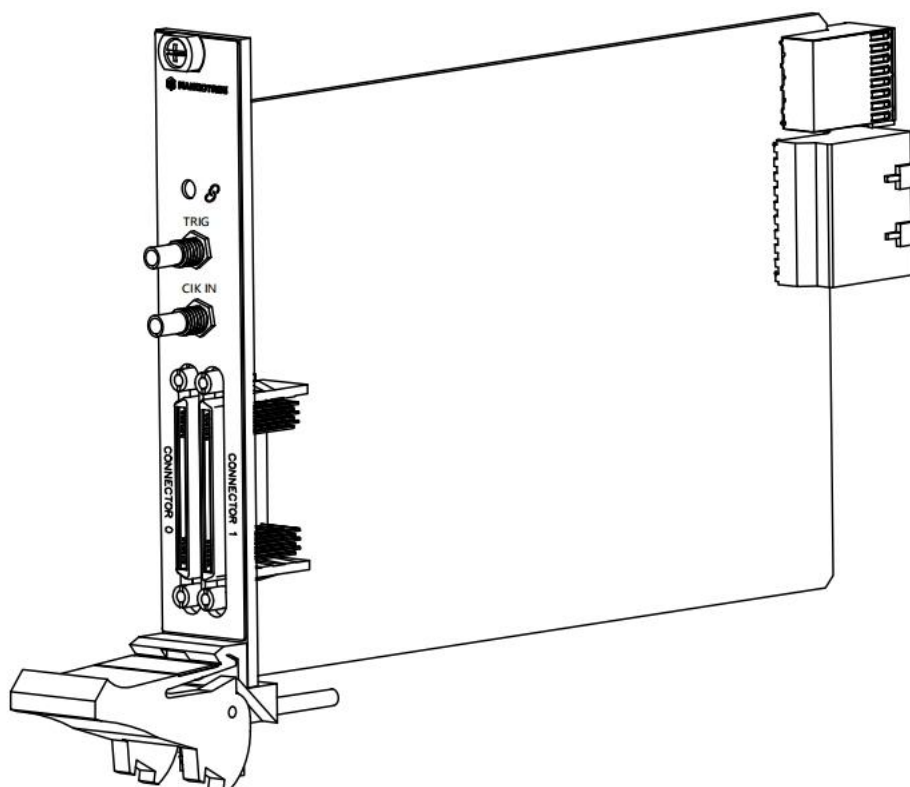
MT-X970

PXle LVDS Digital I/O Module, 800Mbps, 36Chanel

This document contains the specifications for MT-X970. Specifications are typical at 25°C unless otherwise noted.



Caution Using the MT-X970 in a manner not described in this document may impair the protection the MT-X970 provides.



Channel Specifications

Number of connectors	2 SMA(TRIG and CLOCK IN) and 2 VHDCI
Number of digital I/O channels	16 LVDS data in, 16 LVDS data out 4 LVDS PFI and 3 single-ended PFI
Number of clock input terminals	2, CLOCK IN(SMA) and STROBE(VHDCI)
Number of clock output terminals	1,CLK OUT(VHDCI)

Single-Ended Chanel (TRIG, PFI, CLOCK IN)

Maximum data rate	100 Mb/s
Voltage Levels	3.3V

LVDS Channels

Maximum data rate	800 Mb/s	
Generation (Data out, Clock out)		
Offset Voltage	1.125 V	Minimum
	1.2 V	Typical
	1.375V	Maximum
Differential Voltage	247 mV	Minimum
	340 mV	Typical
	454 mV	Maximum

Acquisition (Data in, STROBE)

Magnitude of Differential Input Voltage	0.1 V	Minimum
	0.8 V	Maximum
Input Voltage	0 V	Minimum
	4 V	Maximum

Reconfigurable FPGA

FPGA type	Kintex-7 325T
Number of flip-flops	407,600
Number of LUTs	203,800
Embedded Block RAM	16,020 kbits
Number of DSP48 slices	840

Onboard Memory(FPGA)

DDR3	512MB
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Maximum Power Requirements

Power requirements are dependent on the digital output loads and configuration of the LabVIEW FPGA VI used in your application.

+3.3V	500 mA
+12 V	1.2 A

Physical Characteristics

Weight	300g
Dimensions	See end of this document

Safety Voltages

This product is designed to meet the requirements of the following electrical equipment safety standards for measurement, control, and laboratory use:

- IEC 61010-1, EN 61010-1
- UL 61010-1, CSA C22.2 No. 61010-1

CE Compliance

This product meets the essential requirements of applicable European Directives, as follows:

2014/35/EU; Low-Voltage Directive (safety)

2014/30/EU; Electromagnetic Compatibility Directive (EMC)

2014/34/EU; Potentially Explosive Atmospheres (ATEX)

Shock and Vibration

To meet these specifications, you must panel mount the system.

Operating vibration

Random (IEC 60068-2-64)	5 g _{rms} , 10 Hz to 500 Hz
Sinusoidal (IEC 60068-2-6)	5 g, 10 Hz to 500 Hz
Operating shock (IEC 60068-2-27)	30 g, 11 ms half sine; 50 g, 3 ms half sine; 18 shocks at 6 orientations

Environmental

Refer to the manual for the chassis you are using for more information about meeting these specifications.

Operating temperature	-40 °C to 70 °C (IEC 60068-2-1, IEC 60068-2-2)
Storage temperature	-40 °C to 85 °C (IEC 60068-2-1, IEC 60068-2-2)
Ingress protection	IP40
Operating humidity (IEC 60068-2-78)	10% RH to 90% RH, noncondensing Storage
humidity (IEC 60068-2-78)	5% RH to 95% RH, noncondensing Pollution
Degree	2
Maximum altitude	4,000 m

Indoor use only.

Support

MT-RIO上手指南：

<http://server.mangotree.cn:9900/WebFile/Downloads/上手指南/MT-RIO/>



RIO上手指南

MT-RIO视频教程：

<http://server.mangotree.cn:9900/WebFile/Downloads/视频教程/MT-RIO/>



RIO视频教程

MT-Master上手指南：

<http://server.mangotree.cn:9900/WebFile/Downloads/上手指南/MT-Master/>



Master上手指南

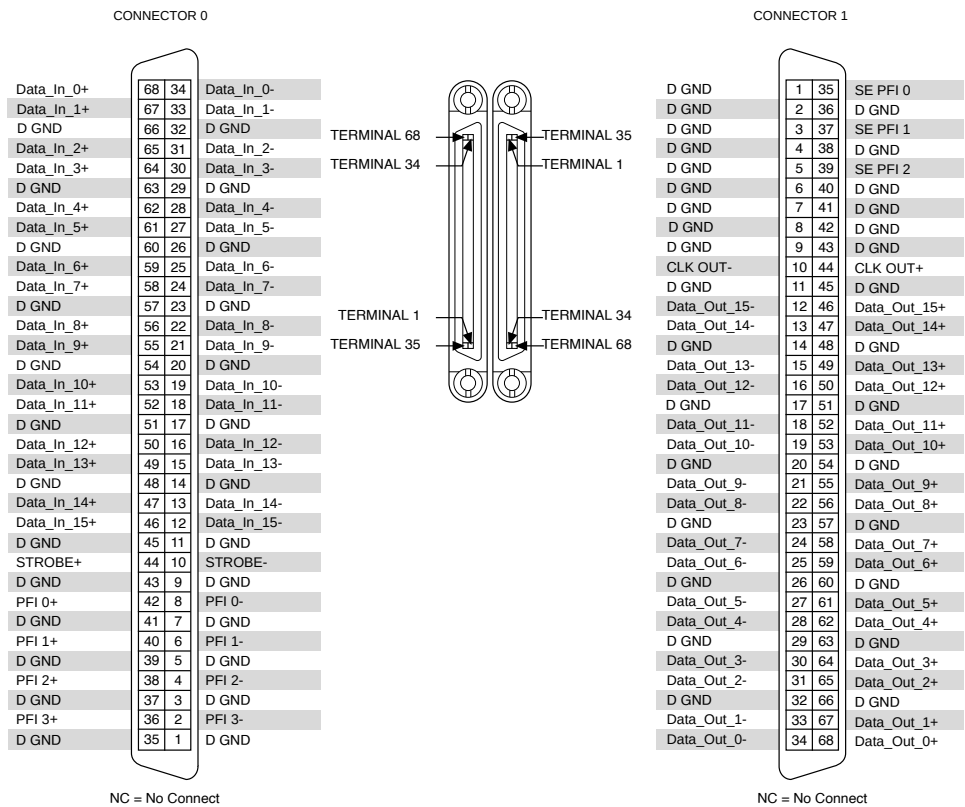
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Master视频教程

MT-X970 Pinout



Dimensions: (mm)

